

Tianyi Xu

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Research Interests

I develop efficient and adaptive methods for AI agents, spanning multi-agent systems, test-time training and adaptation, self-evolution, and data-centric AI, where capability, reliability, and compute efficiency must be jointly optimized.

Education

Oregon State University

Sept 2026 – Present

Ph.D. in Computer Science

Advisor: Huazheng Wang

University of Wisconsin-Madison

Sept 2021 – Dec 2025

B.S. in Computer Science, Data Science, and Mathematics; GPA: 3.918/4.000

Publications

* denotes equal contribution.

Published

[1] **OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration** [arXiv] [Blog] []

ICML 2026
DATA-FM @ ICLR 2026

Shaobo Wang*, Xuan Ouyang*, Tianyi Xu*, Yuzheng Hu, Jialin Liu, Guo Chen, Tianyu Zhang, Junhao Zheng, Yubo Ma, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang

🏆 **ICML Oral (top 0.7%); Top #1 Paper of the Day at Hugging Face Daily**

[2] **EvoPool: Evolutionary Programmatic Annotation for Label-Efficient Specialized Supervision** [arXiv] []

Findings of EMNLP 2026

Tianyi Xu, Yaolun Zhang, Xuan Ouyang, Huazheng Wang

[3] **Stem rot affects the structure of rhizosphere microbiome in berseem clover (*Trifolium alexandrinum*)** [DOI] []

Rhizosphere, 2025

Salma Mukhtar, Zain Ahmad, Noor Khan, Tianyi Xu, Dalaq Aiysha

Under Review

[4] **EVOCHAMBER: Test-Time Co-evolution of Multi-Agent System at Individual, Team, and Population Scales** [arXiv] []

Under Review

Yaolun Zhang*, Tianyi Xu*, Shengyu Dai, Zhenwen Shao, Qingyun Wu, Huazheng Wang

[5] **Generating the Unheard: Phylogeny-Guided Latent Generation for Ancestral Sound Reconstruction**

Under Review

Tianyi Xu, Shrinaath Narasimhan, Evan Gorstein, Santiago Perea, Yunyi Shen, Claudia Solís-Lemus

[6] **MAST: Label-Efficient, Robust, and Generalizable Sound Detection for Biodiversity Monitoring via Masked Audio Pretraining and Self-Training**

Under Review

Tianyi Xu, Daniel L. Pimentel-Alarcón, Zuzana Buřivalová, Claudia Solís-Lemus

[7] **SITA: Learning Speaker-Invariant and Tone-Aware Speech Representations for Low-Resource Tonal Languages** [arXiv] []

Under Review

Tianyi Xu*, Xuan Ouyang*, Binwei Yao, Shoua Xiong, Sara Misurelli, Maichou Lor, Junjie Hu

[8] Combined effects of methyl bromide and soil amendments on soil bacterial and fungal communities in turfgrass [\[Preprint\]](#) [🔗](#)
Tianyi Xu*, Salma Mukhtar*, Evan Gorstein, Claudia Solís-Lemus, Ming Yi Chou, Paul Koch

Under Review

Talks / Presentations

Friday at 4 Seminar – Invited Talk	Dec 2025
<i>“Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring”</i>	
Department of Plant Pathology, University of Wisconsin–Madison	
Undergraduate Research Symposium – Poster Presentation	Apr 2024, Apr 2025
26th & 27th Annual Undergraduate Symposium, University of Wisconsin–Madison	
<i>“Biodiversity Detection Using Acoustic Signals with Deep Learning”</i>	
Presented as part of the Holstrom Environmental Fellowship requirement (2025).	
Abstract (2025, p. 164) 🔗 , Abstract (2024, p. 139) 🔗	

Awards

ICML 2026 Oral (Top 0.7%), Seoul	2026
Phi Beta Kappa (PBK), Inducted	2025
Holstrom Environmental Research Fellowship	2024
Awarded \$3,000 + \$1,000 (advisor) for research on acoustic event detection in tropical rainforests using deep learning.	
UW–Madison Summer Scholarship	2024
Awarded \$1,000 for Summer 2024 studies in recognition of outstanding academic achievement.	
Dean’s List (Six Semesters)	2021–2024
Recognized for maintaining a GPA above 3.85 in Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2024, and Fall 2024.	

Research Experiences

Research Intern, Johnson & Johnson	<i>Remote</i> <i>June 2026 – Sept 2026</i>
○ Researched self-improving multi-agent systems, developing methods for LLM agents to refine their coordination and problem-solving strategies from execution feedback rather than human supervision. ○ Applied reinforcement learning to long-tailed, class-imbalanced classification of medical complaint reports, designing rewards that up-weight rare categories to counter the head-class collapse of standard supervised fine-tuning.	
Research Intern, Epic Lab (SJTU)	<i>Remote</i> <i>Aug 2025 – Feb 2026</i>
<i>Advisor: Linfeng Zhang, Mentor: Shaobo Wang</i>	
○ Co-developed OPUS: an optimizer-aware, efficient, validation-guided data selection method for LLM pre-training with Adam and Muon optimizers. ○ Built the distributed pre-training and continued pre-training stack for Llama3, Qwen3, and GPT-2 using Megatron with FSDP and TP on multi-GPU, and vLLM-based evaluation; +2 to 3 pts avg on downstream tasks.	
Research Assistant	<i>Madison, WI</i> <i>July 2025 – Present</i>
<i>Advisor: Junjie Hu</i>	
○ Led SITA, addressing tone-collapse in multilingual self-supervised speech models for low-resource tonal languages by adapting wav2vec2 and XLS-R into tone-aware, speaker-invariant representations. ○ Designed a multitask training recipe combining contrastive learning, ASR CTC auxiliary loss, and knowledge distillation, implemented in fairseq with distributed multi-GPU training; achieved 0.95 to 0.99	

Top-1 and Top-5 cross-speaker retrieval while preserving tone distinctions.

Research Assistant

Advisor: Claudia Solís-Lemus

Madison, WI

June 2025 – Present

- Led **MAST**, a label-efficient and shift-robust framework for **time-frequency biodiversity sound detection** combining masked audio pretraining, audio-aware ViT adapters, **box-level contrastive learning**, and **iterative self-training**; provided theoretical analysis casting self-training as **distributionally robust optimization** with monotonic improvement guarantees, achieving **+0.22 mAP** and **+0.24 F1** over the strongest baseline under cross-site shift on rainforest soundscapes.
- Led **Generating the Unheard**, the first generative framework for **ancestral sound reconstruction** of bird vocalizations, combining VAE latent encoding, a learned **tree-metric trait projection** aligned with phylogenetic distances, **Brownian ancestral inference**, and a closed-form **anchored inverse lift**; proved row-orthonormality and Brownian motion posterior optimality, validated on Tyrannidae and Paridae clades.